

Information Architecture Design

AU Roadmap Project

Project Name: Adelaide University Program Roadmap

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1. Document Overview

1.1 Purpose of the Document

The Adelaide University Program Roadmap platform requires an information architecture that separates distinct user journeys while keeping the overall experience coherent. This document defines how the platform's content, navigation, labelling, and page hierarchy are structured to support three primary user groups: current students, prospective students, and administrators.

The document serves as a formal design record covering the overall site structure, page hierarchy and content relationships, navigation systems, labelling and terminology conventions, content classification and taxonomy, search and indexing structure, and user flow patterns. It is the primary reference for developers implementing the navigation structure and frontend routing, and it bridges the gap between high-level user requirements and detailed technical implementation.

1.2 Project Background

The AU Roadmap platform is a web-based system designed for two core user groups. Current students access an authenticated portal for roadmap, timetable, support, community, and employability-related information. Prospective students access a public website for program exploration, comparison, career outcomes, campus life information, and application guidance. Administrators manage structured content through a dedicated admin panel.

The platform's information architecture is not a single flat website structure. It is a role-aware, multi-environment architecture that separates public exploration, authenticated student support, and administrative content management into distinct structural spaces. The architecture draws on established principles from Rosenfeld, Morville, and Arango's foundational text on information architecture for the web [1], supplemented by IA Authority guidance on navigation design [2], site maps [3], and labelling systems [4].

1.3 Scope of the Information Architecture

The document covers the top-level site structure, page hierarchy and site map, navigation design, content grouping and labelling, taxonomy and metadata logic, search and indexing structure, and user flow patterns. Implementation details such as specific component code, API endpoint design, and database schema are covered in separate technical documentation.

1.4 Design Objectives

The architecture was designed around four objectives: separate public and authenticated content clearly; support different user goals without mixing unrelated tasks; reduce cognitive load through logical grouping and familiar labelling; and allow future expansion without structural redesign. These objectives align with basic principles of information architecture as outlined by YaleSites [5].

1.5 User-Centred Design Basis

Different user groups hold fundamentally different mental models. Current students think in terms of progression, schedule, support, and next steps. Prospective students think in terms of exploration,

comparison, decision-making, and application. Administrators think in terms of content ownership, structured records, and maintenance tasks. The architecture reflects these mental models by providing each group with an environment organised around its own conceptual structure rather than imposing a single site-wide logic.

2. User Needs and Business Goals

2.1 User Needs Summary

The architecture responds to three distinct user-need profiles, as identified in the User Requirement Analysis Report [6].

Current Students need to understand their program structure, check progression across semesters and years, access timetable information, locate support resources quickly, and connect academic planning with career context.

Prospective Students need to browse programs easily, compare options, understand outcomes and campus experience, and find application guidance without confusion.

Administrators need to manage structured content domains, maintain consistent records, and work within a content hierarchy that mirrors the platform's data model.

2.2 Business and Project Goals

The information architecture supports broader project goals: presenting Adelaide University program information in a structured and navigable format; centring the student experience around the roadmap concept as the organising metaphor; supporting prospective student decision-making through clear exploration pathways; and enabling efficient administrator content management through domain-based organisation. These goals derive from the original Project Proposal [7].

3. Architecture Model

3.1 Top-Level Structure

The platform is organised into four major environment groups, implemented across five route prefixes in the prototype. Public Home and Authentication (`/`, `/login`, `/register`) provides the entry point for all users. Public Exploration (`/explore`) hosts the prospective student experience including program browsing, career outcomes, campus life, and application guidance. Student Portal (`/student`) contains all authenticated student functionality including dashboard, roadmap, timetable, resources, community, and profile. Admin Panel (`/admin`) provides authenticated administrator access for content and user management.

The architecture treats Research as a content section within the public-facing environment rather than a fully independent top-level entity in the route structure. This is reflected in the prototype's routing configuration.

Three-Environment Architecture of the AU Roadmap Platform

Role-aware separation with a unified backend and database supporting all three audiences

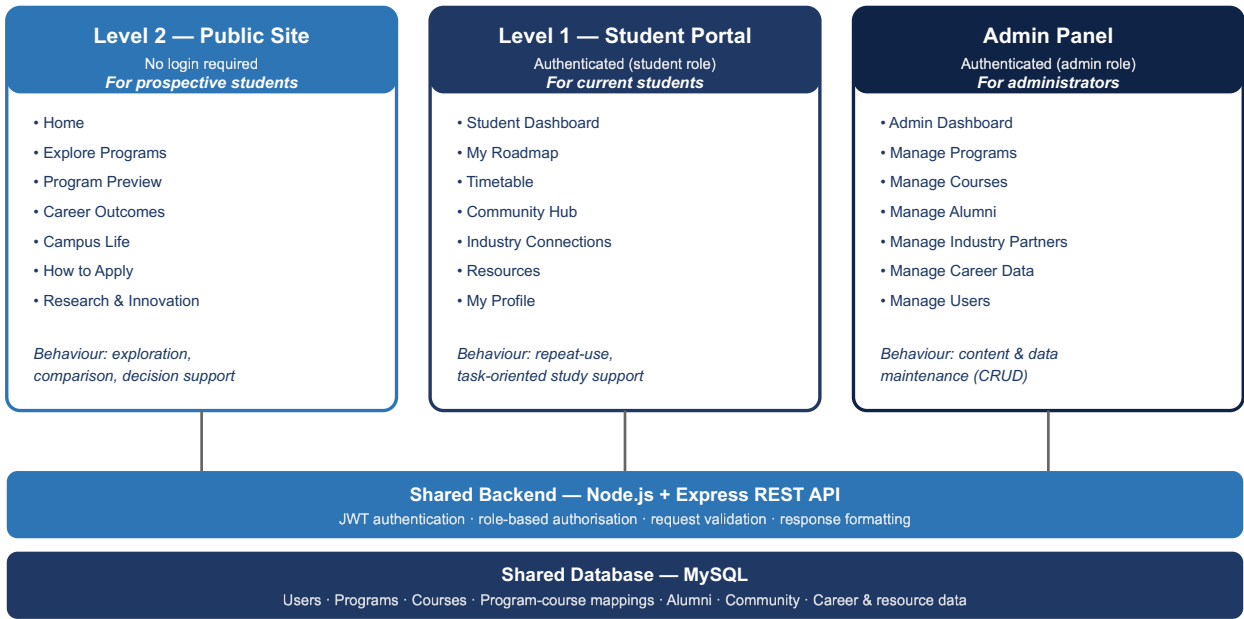


Figure 1. Three-environment architecture of the AU Roadmap platform.

3.2 Structural Logic

The structure uses a hybrid organisational model. At the site level it is hierarchical, with four top-level prefixes defining the major user environments. Within the student portal it is task-based, reflecting the repeat-use patterns of enrolled students. Within the admin panel it is domain-based, mapping directly to content management responsibilities. Within public exploration it is topic-based, supporting the open-ended browsing behaviour of prospective students.

4. Site Map

4.1 Numbered Site Map

The numbered site map is intended to make page relationships easier to reference in design and development discussions. The full numbered site map is provided as a separate figure. This section summarises the structure and hierarchy logic.

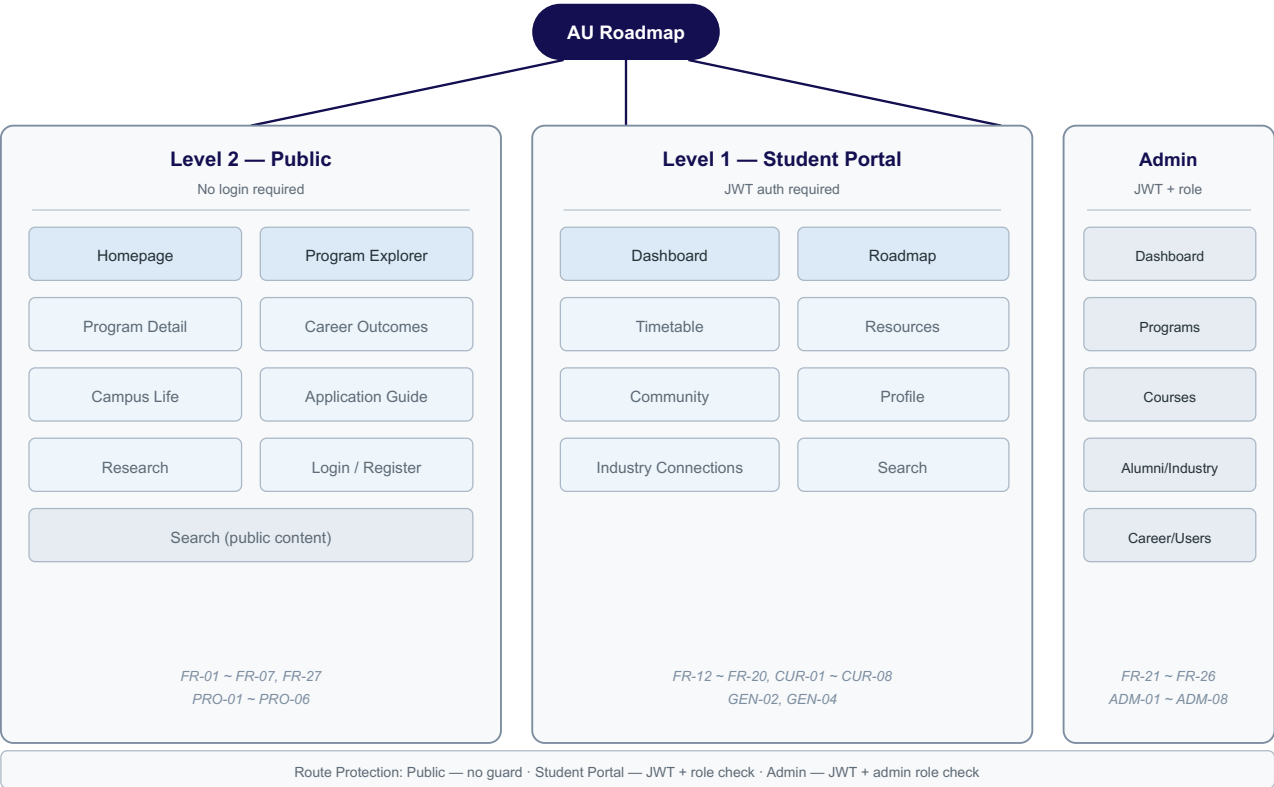


Figure 2. Numbered site map of the AU Roadmap platform.

4.2 Route-Based Structure

The application route structure mirrors the logical grouping described in Section 3. Top-level URL prefixes correspond to environments, and second-level routes correspond to pages within each environment. This route-based organisation ensures that navigation, access control, and content structure remain consistent.

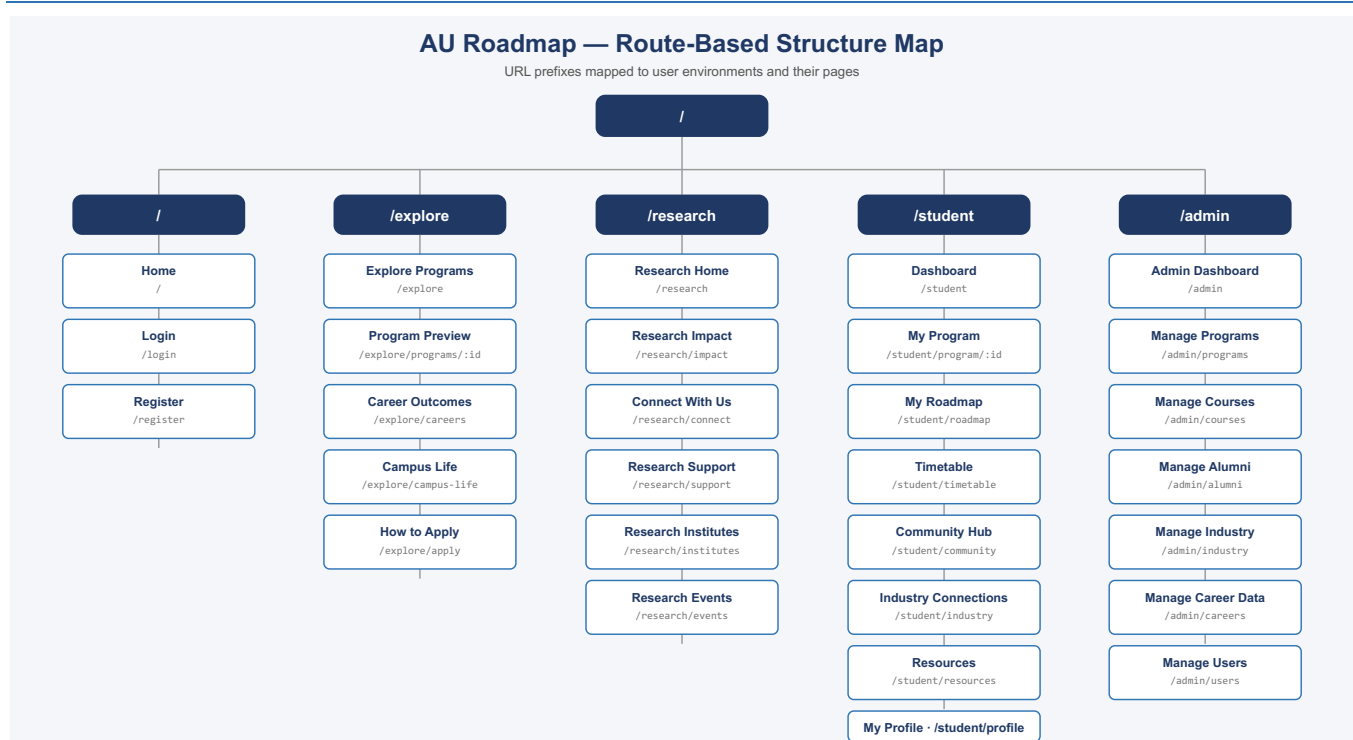


Figure 3. Route-based structure.

4.3 Hierarchy Interpretation

The hierarchy deliberately keeps the top level broad and the lower levels focused. Public users encounter a discovery-oriented structure built around exploration topics. Student users move into a task-oriented portal structure built around their study progression needs. Admin users operate within a content-domain structure mapped to their management responsibilities.

5. Navigation System Design

5.1 Navigation Overview

The platform uses multiple navigation systems rather than a single navigation method, because different parts of the system support fundamentally different user behaviours. The main navigation systems are global navigation, local navigation, utility navigation, contextual navigation, and breadcrumb navigation. This multi-system approach follows established IA practice for complex, multi-audience websites [2].

5.2 Global Navigation

Global navigation applies primarily to the public-facing website. It supports high-level movement between major areas such as Home, Explore Programs, Research, and the Login / Register entry points. In the student portal and admin panel, global navigation is replaced by sidebar-based local navigation, because these environments require task-oriented and domain-oriented navigation respectively.

5.3 Local Navigation

Local navigation is most clearly implemented through sidebar navigation in the student portal and admin

panel.

The student portal uses a sidebar because student behaviour is task-based and repetitive. Students return to the portal frequently for specific actions: checking the roadmap, viewing the timetable, browsing resources, or reading community discussions. A persistent sidebar supports efficient task switching without requiring users to re-orient themselves.

The admin panel also uses sidebar navigation, but its logic is based on content ownership rather than student workflow. Each menu item corresponds to a domain that administrators manage directly: Programs, Courses, Alumni, Industry Partners, Career Data, and Users. This domain-based approach reduces the need for administrators to navigate across unrelated sections.

5.4 Utility Navigation

Utility navigation supports secondary actions: login, register, account access, and return-to-home affordances. In the student portal context, a utility element also provides a path back to the public website, acknowledging that students may wish to explore public-facing content while remaining logged in.

5.5 Contextual Navigation

Contextual navigation appears inside content areas where users move based on related content rather than top-level structure. Typical examples include moving from Explore Programs into a Program Preview page, or from a Program Preview page into Career Outcomes content for that specific program. These cross-links reflect natural information-seeking behaviour rather than formal hierarchy.

5.6 Breadcrumb Navigation

Visible breadcrumbs are not currently implemented as a major interface element in the prototype. However, breadcrumb logic is supported naturally by the route structure and would be especially suitable for the public exploration environment where users may drill several levels deep into program and career content.

5.7 Navigation Design Principles

The navigation system follows five principles: keep primary labels short and familiar; separate broad exploration from repeated task execution; avoid mixing public and authenticated navigation in the same navigation component; use sidebar navigation for task and domain environments; and ensure that the primary path from login to the student's roadmap requires no more than one click.

6. Content Inventory and Classification

6.1 Purpose of the Content Inventory

The content inventory identifies the major content types in the system and the logic used to organise them. This ensures that the architecture is based on stable information objects rather than on temporary page arrangements. The labelling strategy follows guidance from IA Authority on consistent labelling systems [4].

6.2 High-Level Content Inventory

Content Area	Main Content Units	Primary Audience
Public Home	Homepage, Research, Login/Register	Prospective Students, General Public
Explore Programs	Program List, Program Detail, Career Outcomes, Campus Life, How to Apply	Prospective Students
Student Portal	Dashboard, My Roadmap, Timetable, Resources, Community Hub, Industry Connections, My Profile	Current Students
Admin Panel	Dashboard, Program Mgmt, Course Mgmt, Alumni Mgmt, Industry Mgmt, Career Mgmt, User Mgmt	Administrators

Table 1. High-level content inventory across the four major platform environments.

7. Labelling and Terminology

7.1 Purpose of Controlled Labelling

Consistent labelling reduces user confusion and supports learnability. When the same concept uses the same label everywhere in the system, users build a reliable mental model of the information space. The labelling decisions in this architecture reflect IA best practice on labelling systems [4].

7.2 Public-Facing Labels

Label	Content	Rationale
Explore Programs	Program browsing and search	Supports open-ended exploration; more inviting than “Program List”
Career Outcomes	Graduate outcomes per program	Directly addresses the decision-support question prospective students ask
Campus Life	Student experience information	Matches common university marketing terminology known to prospective applicants
How to Apply	Application guidance	Task-oriented label that explicitly answers the question prospective students are asking

Table 2. Public-facing labelling rationale.

7.3 Student Portal Labels

Label	Content	Rationale
Dashboard	Student overview	Standard web application label for a personal landing page with summary information
My Roadmap	Personal study progression structure	Directly reflects the project theme; “My” signals personal ownership and relevance

Timetable	Schedule view	Standard university terminology used consistently across Australian higher education
Community Hub	Discussion and peer interaction area	“Hub” signals a multi-function social space beyond a simple forum
Industry Connections	Employability and partner content	Future-facing label that connects current study with career context
Resources	Support links and services	Broad but familiar label for support content organised by student need categories
My Profile	Account identity area	Clear account label; “My” signals personal ownership consistent with My Roadmap
Research	Research-related institutional content	Short, distinct label that separates research content from program exploration

Table 3. Student portal and public labelling rationale.

7.4 Label Consistency Rules

The architecture follows several labelling consistency rules. The “My” prefix is reserved for personal student-owned spaces (My Roadmap, My Profile) to visually distinguish personal content from general content. Portal sections use substantive nouns (Dashboard, Timetable, Resources) rather than verbs (View Timetable, Browse Resources), consistent with navigation labelling conventions. Admin section labels use the “Manage” prefix where the page supports CRUD operations (Manage Programs, Manage Courses, Manage Users), providing immediate clarity about the action scope.

7.5 Terminology Considerations

Several labels represent deliberate choices over alternatives. “My Roadmap” was selected over “Program Structure” or “Course Matrix” because it is more accessible to students and reinforces the project’s core concept. “Community Hub” was chosen over “Forum” or “Discussion Board” to suggest a broader social and support space rather than a simple message board.

8. Taxonomy and Metadata System

8.1 Purpose of Taxonomy

Taxonomy helps classify content in ways that improve browsing, filtering, search, and management. In the AU Roadmap platform, the taxonomy is practical and lightweight, because the content volume is moderate and the primary navigation relies more on structural hierarchy than on faceted classification. However, taxonomy still plays an important role in organising student resources by need category and supporting program filtering during exploration.

8.2 Taxonomy Categories

Domain	Classification	Purpose
Programs	Level (undergraduate / postgraduate / research), faculty, program code	Supports filtering during program exploration; structures preview

		information; underpins admin content management
Student Resources	Academic, career, wellbeing, financial, technology, library	Organises support content by student need rather than by institutional department, matching how students think about support
Community	General, academic, career, social	Improves discussion readability and supports future filtering within the Community Hub
Career Data	Career paths, graduate outcomes, salary trends, employment rates	Reflects the decision-making questions prospective students are most likely to ask when comparing programs

Table 4. Taxonomy categories across major content domains.

8.3 Metadata Considerations

Although the prototype does not expose a large explicit metadata editor, the architecture assumes that content can be described using structured metadata including title, category, relevant program codes, content type, and last-updated date. This metadata supports search indexing, content filtering, and future administrative workflows.

9. Search and Indexing System

9.1 Search Role in the Architecture

Search serves as a secondary navigation mechanism, supporting users who prefer to locate content directly rather than browsing hierarchically. In the AU Roadmap platform, search complements the primary browsing-based structure by providing a fallback path for users who know what they are looking for or who have not yet formed a clear browsing strategy.

9.2 Search Scope

The system supports unified search across programs, courses, alumni, industry partners, and career-related content. This cross-domain scope improves content discoverability and reduces the need for users to know which section contains the information they need.

9.3 Search Placement and Navigation Logic

Search is implemented as a tool-navigation element (a search bar in the header) rather than as a standalone site-structure page. This is appropriate because browsing remains the primary navigation method in the public exploration environment, while task repetition dominates in the student portal. Search is available in both environments but does not compete with the primary navigation systems.

9.4 Indexing Considerations

For search to function meaningfully, content should be indexable by name or title, program code, content category, content type, and related domain. The architecture already structures content along these dimensions, making search indexing a natural extension of the existing organisational logic.

10. User Flows

10.1 Purpose of User Flows in IA

User flows are included because information architecture is not only about static hierarchy; it is also about how users move through the system to complete tasks. The flows below summarise the most common pathways identified in the requirement analysis and validated against the platform's navigation design.

10.2 Prospective Student Flow

A typical prospective-student journey reflects exploratory behaviour followed by evaluation and decision support. Users enter through the Home page, browse programs on Explore Programs, preview individual programs, and branch into supporting content such as Career Outcomes, Campus Life, and How to Apply.

```
Home → Explore Programs → Program Preview
├→ Career Outcomes
├→ Campus Life
└→ How to Apply
```

10.3 Current Student Flow

A typical current-student journey reflects repeat-use academic support behaviour. After login, the Dashboard provides an orientation point with summary context. From there, students most commonly access the Roadmap for progression planning, followed by Timetable, Community Hub, and Resources as needed.

```
Login → Student Dashboard
├→ My Roadmap
├→ Timetable
├→ Community Hub
└→ Resources
```

10.4 Administrator Flow

A typical administrator journey reflects task-based content maintenance. After login, the Admin Dashboard provides an overview with summary statistics. Administrators then select a content domain and perform CRUD operations within that domain.

```
Login → Admin Dashboard
├→ Manage Programs → Create / Update / Delete
├→ Manage Courses → Create / Update / Delete
└→ Manage Users → View / Update / Delete
```

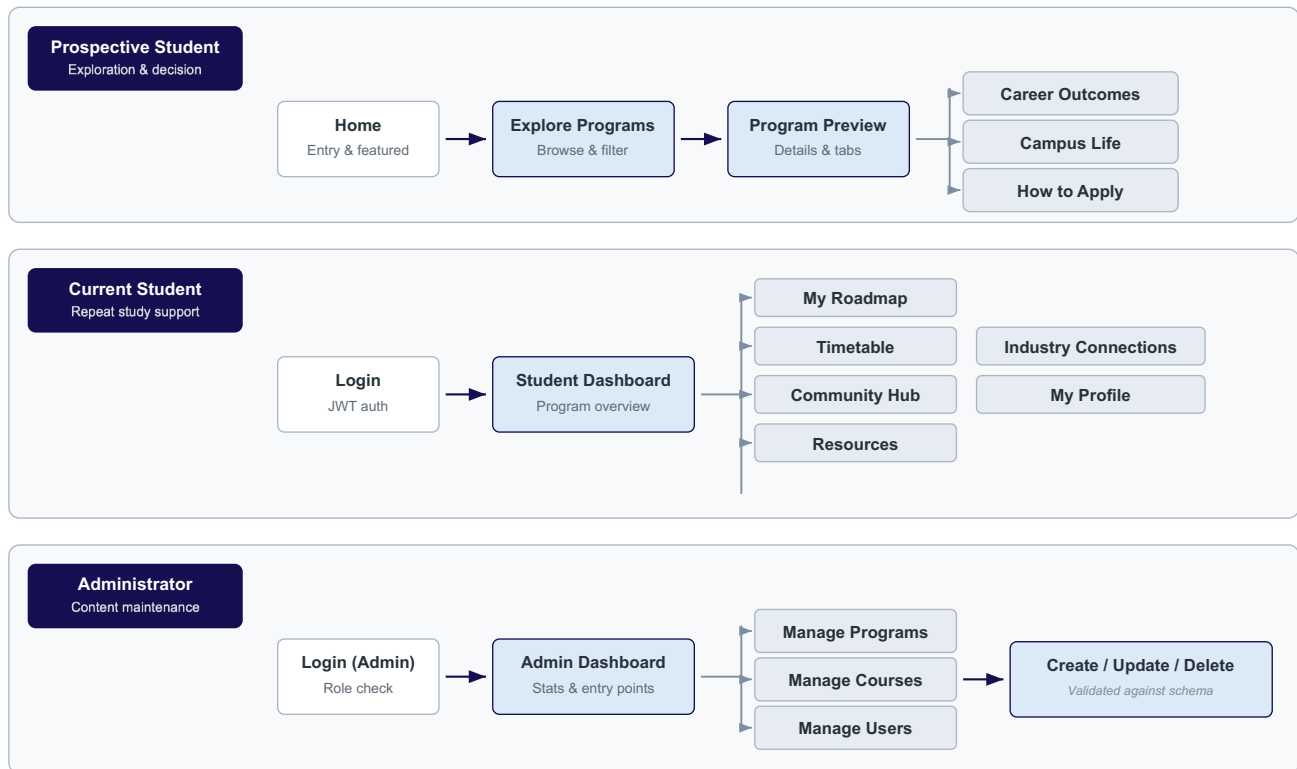


Figure 4. Primary user flows across the platform for prospective students, current students, and administrators.

11. Cross-Linking and Non-Linear Navigation

Although the architecture uses a strong hierarchical structure, it also supports non-linear movement where user behaviour demands it. Users do not always move in a strictly top-down manner, and the architecture accommodates cross-links between related content areas.

Key cross-link relationships include: the Dashboard linking to My Roadmap as the primary next action; Explore Programs linking directly into Program Preview pages; Program Preview pages linking to Career Outcomes content for the same program; and the Community Hub linking to relevant Resources and Industry Connections.

These cross-links are lightweight and do not require a separate navigation component. They appear as contextual links within page content, preserving the clarity of the primary navigation systems while supporting natural information-seeking behaviour. This approach is consistent with established site map and hierarchy practices [3].

12. Structural Rationale

Four key design decisions shaped the final architecture. Rather than treating each as an isolated choice, this section explains the reasoning behind them as a coherent set of structural decisions.

12.1 Multiple Environments vs. Single Website

The project proposal's two-level concept strongly influenced the architecture: Level 1 as a student assistance portal, Level 2 as a public-facing decision-support website. The subsequent addition of an admin environment created a three-environment model. The alternative of a single-site structure would have mixed unrelated user tasks and made access control more complex. The multi-environment approach keeps each user group's experience focused and avoids cross-contamination between exploration, task execution, and content management.

12.2 Roadmap as the Student Portal Anchor

The project is fundamentally about program roadmaps. Translating this into architecture means the roadmap is not one feature among many but the organising centre of the Level 1 experience. The Dashboard leads to the Roadmap; the Timetable and Resources gain meaning from their relationship to roadmap-oriented planning. This anchor concept is unique to this project and distinguishes it from generic student portal designs.

12.3 Exploration as the Public Experience Model

Prospective students begin with uncertainty. They may not know which program they want, what career it leads to, or whether the campus environment suits them. The public architecture therefore supports broad exploration first, followed by progressive narrowing through program previews, career outcomes, and campus information. This is fundamentally different from the task-execution model used in the student portal.

12.4 Domain-Based Admin Organisation

Administrators approach the system through content ownership: which program record, course entry, or alumni profile needs updating. A domain-based admin structure — where each sidebar section corresponds to a distinct content domain — matches this mental model directly and avoids the inefficiency of navigating through unrelated pages to reach management functions.

13. Scalability and Future Extension

The current architecture is designed to remain extensible. Because the platform uses a section-based, route-grouped structure, future content areas can be added under existing top-level prefixes without disrupting the core hierarchy.

Specific extensions the architecture can accommodate include: program comparison tools within the public exploration environment; richer account management within the student portal; advanced content scheduling for administrators; integration with live university data feeds; and mobile-responsive refinements to navigation patterns.

14. Conclusion

The information architecture of the Adelaide University Program Roadmap platform provides a clear, scalable structure for three major user contexts: public exploration, current student support, and administrative content management. Each environment uses navigation patterns, labelling conventions,

and content organisation appropriate to its users' mental models and task patterns.

The architecture reflects the project's central idea: users should be able to find and understand program-related information more easily, whether they are currently enrolled or still considering enrolment. The separation of environments, the roadmap-anchored student portal, the exploration-based public experience, and the domain-based admin structure all serve this central purpose.

References

- [1] L. Rosenfeld, P. Morville, and J. Arango, Information Architecture: For the Web and Beyond, 4th ed. Sebastopol, CA, USA: O'Reilly Media, 2015.
- [2] Information Architecture Authority, "Navigation Design." [Online]. Available: <https://informationarchitectureauthority.com/navigation-design>
- [3] Information Architecture Authority, "Site Maps and Hierarchies." [Online]. Available: <https://informationarchitectureauthority.com/site-maps-and-hierarchies/>
- [4] Information Architecture Authority, "Labeling Systems." [Online]. Available: <https://informationarchitectureauthority.com/labeling-systems>
- [5] YaleSites, "Basic Principles of Information Architecture." [Online]. Available: <https://yalesites.yale.edu/explore-resources/basic-principles-of-information-architecture>
- [6] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "AU Roadmap User Requirement Analysis Report," Group C261W-4037, Industry Research Project, Adelaide University, May 10, 2026.
- [7] J. Liu, Y. Huang, Y. Peng, and V. Thanki, "Development of Program Roadmap — Project Proposal," Group C261W-4037, Industry Research Project, Adelaide University, Mar. 18, 2026.